Akash Prajapati

Python Backend Engineer | Django, FastAPI & Microservices

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PROFESSIONAL SUMMARY

Python Backend Engineer with 3.4+ years of experience building scalable backend systems, REST APIs, and microservices using Python, Django, Django REST Framework, and FastAPI. Experienced in PostgreSQL, Redis, Celery, Kafka, database optimization, asynchronous processing, authentication, and third-party integrations across enterprise automotive and e-commerce applications.

TECHNICAL SKILLS

Programming Languages: Python, SQL

Backend Development: Django, Django REST Framework, FastAPI, REST APIs, Microservices

Databases & Caching: PostgreSQL, Redis, Django ORM, Database Design, Query Optimization

Messaging & Asynchronous Processing: Apache Kafka, Celery

Authentication & Integrations: Keycloak, JWT, OAuth 2.0, OTP Verification, RBAC, Payment Gateway Integration, Third-Party API Integration

Cloud & DevOps: AWS S3, RDS, EC2, ELB, Docker, Kubernetes (Fundamentals)

Tools & Concepts: Git, GitHub, Grafana, Swagger/OpenAPI, OOP, System Design, Agile Methodology

Additional Knowledge: LLMs, RAG, OpenAI APIs, Semantic Search, LangChain, LangGraph

PROFESSIONAL EXPERIENCE

Vishleshan Software Solutions Pvt. Ltd. | Software Developer 25th November 2025 – 24th September 2026

Edukaan — Commercial Vehicle Spare Parts E-Commerce

- Developed 100+ REST APIs using Python, Django, and Django REST Framework for core e-commerce workflows on a platform handling 1,000+ orders daily.
- Implemented Smart Search using separate image and text embedding pipelines, enabling product discovery through images, part numbers, product names, categories, and voice queries, returning the top five similarity-based results.
- Developed FastAPI notification services integrated with Apache Kafka, supporting approximately 1,000–10,000 notifications daily.
- Implemented Redis caching for Order History APIs and optimised Django ORM queries to reduce repeated database access and improve response times.
- Built asynchronous report generation and export workflows using Celery and Redis, integrating AWS S3 for file storage and secure presigned downloads.
- Developed DSR/MSR field-representative business workflows and integrated Shiprocket for shipment creation and tracking, and SAP Grow with internal backend services.

Bhasasetu — Hindi Voice-Enabled AI Assistant

- Developed backend APIs and an end-to-end RAG pipeline using Python and Django to deliver context-grounded responses from a product knowledge base.
- Implemented document processing, recursive chunking (500 words per chunk, 20-word overlap), embedding generation using Sentence Transformers, and ChromaDB vector storage for approximately 100–500 knowledge-base items.
- Built semantic retrieval using cosine similarity and dynamic thresholding, with query rephrasing and fallback handling; integrated Whisper, GPT-4, and ElevenLabs APIs for speech-to-text, response generation, and text-to-speech.
- Evaluated approximately 5,000 test cases against expected answers using LLM-based evaluation, achieving an average response accuracy of approximately 87%.

PV Shopee — Tata Motors | Passenger Vehicle Spare Parts E-Commerce

- Integrated Keycloak authentication APIs for login and token-generation workflows using JWT and OAuth 2.0.
- Implemented CAPTCHA validation and supported backend maintenance and production issue resolution.

Sankey Solutions | Solution Analyst 1st June 2023 – 14th November 2025

FleetEdge — Tata Motors | IoT-Based Fleet Management Platform

- Developed backend features across vehicle, wrapper, subscription, and payment services within an approximately 18-microservice fleet management platform using Python, Django, and PostgreSQL.

- Built the Vehicle Day-wise Tracker backend end-to-end, processing historical GPS coordinates, calculating distances using the Haversine formula, identifying trips and stops, and preparing route data for Google Maps visualization.
- Improved API performance by 20–30% by identifying and resolving N+1 query issues and optimising database access using Django ORM Subquery and related query optimisations.
- Implemented role- and subscription-based access controls, managing Fleet Owner and Fleet Manager permissions, site-level vehicle visibility, and plan-based access to vehicle data and fleet management modules.
- Developed Mileage Saarathi APIs and business logic to process driving parameters such as speed, idling, harsh acceleration, braking, and gear shifting/RPM to generate mileage-improvement guidance.
- Implemented historical telemetry synchronisation through Sahaj APIs using 30-minute cron scheduling, checkpoint-based incremental fetching, backlog recovery, and PostgreSQL upserts keyed by VIN and timestamp.
- Integrated CCAvenue payment gateway with request encryption, response decryption, payment callbacks, status updates, subscription synchronisation, and Redis cache updates; also developed vehicle subscription workflows for plan purchases, upgrades, renewals, and expiry.

IQMS — Intelligent Quality Management System | Tata Motors Passenger Vehicle Manufacturing

- Developed the User Portal backend using Python, Django, and PostgreSQL, covering database design and implementation of 50+ REST APIs to digitise paper-based vehicle assembly-line quality inspections.
- Implemented Quality Maker and Quality Checker workflows for inspection recording, defect reporting, corrective actions, and verification across assembly-line operations.
- Developed an Apache Kafka consumer to receive VIN and vehicle information from manufacturing sites and persist incoming data in PostgreSQL; integrated Swagger/OpenAPI documentation for API testing and frontend coordination.

Vidura — AI-Assisted Git Repository Change & Security Analysis

- Independently developed a code analysis solution integrating GitHub APIs to retrieve commit differences and surrounding code context.
- Implemented separate OpenAI API workflows for code-change summarisation and potential security vulnerability detection.
- Developed checks for exposed API keys, access tokens, passwords, email credentials, and AWS access details, generating structured reports with issue descriptions and severity levels.

EDUCATION

Bachelor of Engineering — Computer Engineering
Smt. Indira Gandhi Engineering College, University of Mumbai | Aug 2019 – Jun 2023 | CGPA: 8.88/10

ACHIEVEMENTS

- **Shining Star:** Recognized for outstanding technical contributions.
- **Bar Raiser:** Recognized for problem-solving, technical guidance, and supporting team members.

CERTIFICATIONS

- Machine Learning with Google Cloud
- Android Development

PUBLICATION

- Crop Protection Against Birds Using Deep Learning and IoT, IRJET, Vol. 10, Issue 4, April 2023.